A cached-head 194-state Turing machine for the consistency of PA

Research, construction and verification
prepared for Vladimir Reshetnikov

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Abstract

No verified construction below 194 states was obtained in the searches reported here. The requested alternative is achieved: an explicit binary, single-tape, blank-start machine M_{194C} with **194 working states** and a materially different command protocol from both supplied positive 194-state designs. It combines the successful 67-reader, seven-fold decoder of PA194 with the cached-head Shift/modus-ponens controller of the supplied 195C construction. The resulting count is $134 + 67 - 7 = 194$.

Six native transition rows change. The source has 3,649 primitive instruction sites, compared with 3,673 for PA194. Two implementations independently verify all 4,749 source outcomes for arbitrary natural register values. The second rederives four generic list-sweep lemmas from the literal machine rows. A complete blank-tape calculation reaches the exact source seed after 57,922 transitions. The finite source map agrees exactly with the supplied 195C map; all seven history-sensitive folds remain valid simultaneously.

Together with the retained numerical-calculus calibration and the explicit cached-command proof compiler, these give the ordinary mathematical result $\mathrm{PA} \vdash (\neg\,\mathrm{Halt}(M_{194C}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA}))$. This is not a claim of a newly compiled end-to-end Lean or Rocq theorem. An incremental census covers all 34 incoming archives at the pinned revision, and proves that the new graph is absent even up to arbitrary state renaming and global tape reflection. The package contains the full table, deterministic builders, independent checks, proof references, receipts and search artifacts.

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1 Result, conventions and scope

The main table is `tables/pa194C.tm`. Its working controls are $0, \ldots, 193$, its initial control is 0, and its separate terminal label is 194. The initial head is at cell zero on the all-zero two-way infinite tape. Every transition reads and writes a bit and moves exactly one cell left or right. There are 388 rows, exactly two for each working control, and no rows for the halt label. Initialization is included in the count.

The native table starts at 74 and halts at 254. The dense-to-native list is an exact bijection that conjugates every transition. All 194 working controls are reachable in the pointed transition graph; this statement does not assert that every control is visited during the blank computation.

Achieved alternative: 194 states with a changed proof-search protocol.

$$\mathrm{PA} \vdash (\neg\,\mathrm{Halt}(M_{194C}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})).$$

No sub-194 machine and no global lower bound are claimed.

The dense-table SHA-256 is

`d759313c03b515293fe06b12769fa1771d3829a2bad89febaa9b4391990f3a4b`.

The native-table SHA-256 is

`b35d80b01ce9c2e95c8b29222881f40928247cdeffac322e4643f4f6694cd33c`.

These hashes, the complete table in Appendix A, and the builder identify the literal object independently of descriptive names.

Three proof levels must be kept separate. The new composition has finite and symbolic executable certificates. Its consistency conclusion is an ordinary mathematical proof obtained by explicit simulation and composition with the supplied syntactic PA calibration. A fully kernel-checked arithmetic endpoint for this new table is not supplied. The repository's open iPRA obligations are a further formalization target; they are not the statement requested here.

2 Comparison of the supplied corpus

2.1 Pinned snapshot and complete incremental inventory

The comparison is pinned to ProveIt commit

`b88235336c63a247852fe8ec7f931d51f953632d`.

The four requested areas are PA research, incoming archives, ZFC research and ConciseIPRA [1, 2, 4, 5]. The PA194 packet contains a recursive census at the earlier commit `5ed5a41fa5932433111eea096311364c6087e8e3`. The exact GitHub commit-range comparison establishes that all 32 earlier incoming archives and all direct PA `.tm`/`.txt` machine files are unchanged. There are exactly two net new archive entries: `pa194_conpa_2026-11-13.zip` and `pa195_followup_research.zip`. Both additions were freshly parsed and classified here. Thus the current inventory is an exact incremental extension, not an assertion that the entire earlier computation was rerun.

The resulting inventory has 34 top-level archives, 47 distinct ZIP payloads, 75 archive references, 25,742 member occurrences, 2,473 recognized complete numeric binary-table occurrences, 594 byte forms, and

430 root-free graph classes. Two disconnected diagnostic byte forms cannot be assigned a full-reachability root in this canonicalization. The numbers count ancestors, experiments and rejected machines as well as positive constructions.

Recognition covers complete numeric binary `.tm` and `.txt` members with a unique undefined terminal target. It does not claim to decode every possible table embedded in arbitrary source code or JSON. Manuscript and source reviews supplement this precise numeric-file census. At 194 states there are five byte forms in three graph classes: two positive designs and one explicitly rejected scanner candidate. At 195 there are 17 byte forms in 11 classes; a supplementary search table is not promoted to a consistency construction merely because it has that count.

The inherited census, exact commit comparison, current listing, added ZIPs, new table bytes and incremental program are included. An offline replay is `python3 -B comparison/replay_census.py`. The current full inventory is reconstructed from the frozen baseline plus its exact delta; the positive frontier is distinguished below.

2.2 The positive 194/195-state frontier

Family	Source/controller	Count decomposition	Sites	Startup
Original PA195	Bundled diagonal gate; old controller	$134 + 68 - 7$	3,700	58,154
195S	Separate diagonal head; old controller	$134 + 67 - 6$	3,673	58,154
195C	Separate head; cached-head controller	$134 + 67 - 6$	3,649	57,922
M195S8	Bundled gate; extended scanner	$135 + 68 - 8$	3,700	59,164
PA194	Separate head; new decoder	$134 + 67 - 7$	3,673	58,154
PA194_AD	Same, physical A/D exchange	$134 + 67 - 7$	3,673	58,122
PA194C	**New decoder plus cached-head controller**	$\mathbf{134 + 67 - 7}$	**3,649**	**57,922**

Startup figures are exact blank initialization lengths. They do not establish a uniform runtime ordering of the ensuing infinite proof searches. The PA directories named quotient, dense, all-tape, source-entry, runtime, symbolic and arithmetic endpoint contain different verification layers, not that many additional machine designs.

2.3 Larger designs and rejected reductions

James Chang's PA209 and the U209 ancestor of the U208 packets are distinct machines despite the common state count. U208 demonstrates previous-bit control recovery and several alternative helper routings. PA196 changes the quantifier coding, axiom program and allocation, with an original 3,812-site source. Its six-fold saving is already included in 196; it cannot be subtracted again from its descendants.

The later numerical-calculus improvement replaces an equality-vacuity interface by diagonal vacuity, reducing the emitter to 191 macro tokens. Separating the diagonal clause from the guarded bundle gives 190 tokens and a ten-entry AX stack interface. PA194 optimizes the binary association and sharing of that emitter. The present result combines this decoder with an independently established command-store improvement. These are separate optimization layers, and their compatibility is checked on the complete table.

A shorter source is not automatically a smaller physical machine. For example, the retained direct-negation experiment has 3,441 primitive sites but a verified 197-state implementation. Conversely, the extra command simplification in PA194C changes the protocol and shortens the source without lowering the count.

The latest rejected 194-state candidate is particularly instructive. It has $134 + 68 - 8 = 194$ controls,

a correct source graph and locally valid folds, but a seven-zero source address is confused with the separator by the old scanner. Its first failed raw symbolic outcome is site 1003, the zero branch of logical register 1/physical register 2, after 1,309 preceding outcomes succeeded. Adding a scanner control repairs it to M195S8 at 195. Local folds and source identity alone therefore do not establish physical correctness.

Older diagnostic 195-state classes include two-rotation and warm/extended layouts with respectively 93, 714 and 777 failed primitive contracts; a scanner regrouping; a short-scanner deletion with 96 unrestricted failures; and a zero-jump/helper fold that changes the wrong register and successor. Some rejected 196-state Shift tables have the stronger counterevidence of an actual blank halt after 43,563 steps. These tables are retained as negative controls, not counted as positive consistency constructions.

3 The exact new composition

Start with the supplied PA194 native table. Keep its axiom grammar, physical register map and seven decoder folds. Change precisely the six rows below. A triple w, D, q' means write w, move in direction D, enter q'.

Native control	Read	PA194 action	PA194C action
38	0	$0, R, 28$	$0, R, 11$
38	1	$1, R, 20$	$1, R, 122$
164	1	$1, R, 98$	$1, R, 130$
214	0	$0, R, 188$	$0, R, 312$
214	1	$1, R, 312$	$1, R, 20$
312	0	$0, R, 11$	$0, R, 188$

These are the cached-controller changes from supplied 195C, expressed in the native labels of the common compiler. They are not changes to the arithmetic register engine. The builder verifies the native translation of their original dense PA196 coordinates rather than relying on informal label matching.

At source level the changes are exactly

$$\texttt{node163} = (\texttt{node6}, \texttt{node147}),$$
$$\texttt{node170} = ((\texttt{node36}, (\texttt{node167}, \texttt{node145})), \texttt{node57}).$$

The new emitter grammar is exactly the successful PA194 grammar. A complete ordinal-by-ordinal comparison establishes equality of the new 3,649-site register instruction map with the published 195C map, including every increment, both decrement branches, every structured jump and halt. Flattened word equality alone would not suffice because structured jumps depend on their binary ancestors.

The raw allocation has 134 fixed controls and 67 readers. The unchanged seven fold identifications are

$$2016 \mapsto 2052, \quad 2018 \mapsto 2027, \quad 2022 \mapsto 2032, \quad 2024 \mapsto 2046,$$
$$2037 \mapsto 2015, \quad 2055 \mapsto 2019, \quad 2060 \mapsto 2014.$$

They remove seven controls from the 201-control raw machine. The resulting 194-control table is checked directly, with all folds applied simultaneously. The builder also reconstructs the literal 195C parent from its earlier grammar and six folds, and compares that parent byte-for-byte with the supplied table.

Lemma 3.1 (Local quotient)**.** *The new raw 201-control table and PA194C have equivalent halting behavior from native control 74 on every initial tape. Each raw transition is simulated by a positive segment of at most five folded transitions.*

Proof. There are 23 raw incoming rows into the exceptional fold controls. Every incoming edge to a removed member writes zero and moves right; every incoming edge to its retained member writes one and moves right. Thus the previous bit records the missing control distinction. The start is not exceptional. For each of the seven pairs, both histories and both current bits give four cases. All 28 cases are evaluated in the final simultaneously folded table, with exact complete two-cell tape, head displacement and projected destination. Every intermediate visit stays in those two cells, so arbitrary exterior tape is preserved. All other rows commute with the state projection.

Induction concatenates positive simulations of raw runs. If a folded run first halted after T steps while the first $T+1$ raw steps did not halt, their positive simulations would give a longer nonhalting folded prefix, contradicting determinism. This proves both halting directions and is arithmetizable in PA for coded tapes and traces. □

This quotient theorem concerns the newly compiled raw parent. Preservation from 195C to PA194C additionally uses their common source and the complete physical frame contracts in Section 6.

4 Why the cached-head protocol is materially different

Let $p(a,b) = (a+b)(a+b+1)/2 + b$ be Cantor pairing, and write $[a;w] = p(a,w)$ for a numerical stack. Its total projections satisfy $\mathrm{hd}[a;w] = a$ and $\mathrm{tl}[a;w] = w$, including $[0;0] = 0$. Write a command store as $\langle t, w, d, h\rangle$: current formula, stack, offset and cached formula. Put $I(a,b) = 4p(a,b)$.

The cached Shift is

$$\mathsf{S}\langle t,w,d,h\rangle = \begin{cases} \mathsf{halt}, & h = 1, \\ \langle h, \mathrm{tl}(w), d, \mathrm{hd}(w)\rangle, & h \neq 1. \end{cases}$$

For $y = d + x$, cached modus ponens is

$$\mathsf{MP}_x\langle t,w,d,h\rangle = \begin{cases} \langle 0, w, 0, y\rangle, & I(t,y) = h, \\ \langle 0, w, y, 0\rangle, & I(t,y) \neq h. \end{cases}$$

Successful MP leaves its conclusion in the cache and does not push a stack entry. Shift tests that cache alone. The old PA194 command behavior uses the old Shift/MP protocol, including the $t + h = 1$ Shift test. Consequently the new machine is not merely a relabeling or a physical A/D allocation variant.

For an explicit behavioral distinction, a source boundary with $t = 1, h = 0$ triggers the old Shift test, whereas cached Shift does not halt there. This example distinguishes the command contracts on arbitrary stores; it is not a claim that such a store occurs on a consistent blank computation. The two searches have the same refutability calibration, not identical command-by-command trajectories.

4.1 AX, stack identities and a proof compiler

AX retains the separate diagonal head. It pushes ten entries, including the saved top. With $D = d + x$, $H = h + y$, and $A = c$,

$$\mathsf{AX}_{c,x,y}\langle t,w,d,h\rangle = \langle K(H,D,c), [q_1,\ldots,q_9,t;w], 0, 0\rangle,$$

where

$$(q_1, \dots, q_9) = (\Gamma(A), L_6, L_5, L_4, L_3, L_2, L_1, \rho_8(D, A, H, d), U(d, h)).$$

The L_i are the six logical heads, ρ_8 is the dominated implicational head, and K is the guarded old separation gate. All generated heads except $U(d, h)$ have residue zero modulo four, and $U(d, h)$ has residue two. None is the contradiction code one.

Define words, executed left to right,

$$\begin{aligned} \mathsf{Push} &= \mathsf{AX}_{0,0,0}\,\mathsf{S}^9\,\mathsf{MP}_0, \\ \mathsf{Pop} &= \mathsf{AX}_{0,0,0}\,\mathsf{S}^{11}, \\ \mathsf{Discard} &= \mathsf{MP}_1\,\mathsf{AX}_{1,0,0}\,\mathsf{S}^{10}\,\mathsf{MP}_0. \end{aligned}$$

Their complete contracts are

$$\begin{aligned} \mathsf{Push}\langle t, w, d, h\rangle &= \langle 0, [t; w], 0, 0\rangle, \\ \mathsf{Pop}\langle t, [a; w], d, h\rangle &= \begin{cases} \mathsf{halt}, & t = 1, \\ \langle t, w, 0, a\rangle, & t \neq 1, \end{cases} \\ \mathsf{Discard}\langle t, w, 0, 0\rangle &= \langle 0, w, 0, 0\rangle. \end{aligned}$$

For example, after AXS^{10} the store is $\langle U(d, h), w, 0, t\rangle$. The next Shift tests exactly the saved top. The preceding shifts test only zero or generated heads, which cannot be one. The Push MP fails by its residue mismatch. The two MP failures in Discard follow from $I(t, 1) > 0$ and $I(U(1, 0), 0) > 0$. These arguments also cover the zero-stack alias; no formal stack-length invariant is assumed.

Let Offset_0 be empty and $\mathsf{Offset}_i = \mathsf{MP}_i$ for $i > 0$. Then generalization is realized by

$$\mathsf{Gen}_i = \mathsf{Push}\,\mathsf{Offset}_i\,\mathsf{S}\,\mathsf{AX}_{i,0,0}\,\mathsf{S}^{10}, \quad \mathsf{Gen}_i\langle a, w, 0, 0\rangle = \langle U(i, a), w, 0, 0\rangle.$$

Ordinary axiom head q_j, $1 \le j \le 7$, is selected by

$$\mathsf{Discard}\,\mathsf{AX}_{A,D,H}\,\mathsf{S}^{j+1}\,\mathsf{Pop}^{9-j}.$$

The dominated head and guarded gate have the explicit selectors retained in `reference/cached/composite_command_audit.tex`. They return the chosen formula with the original stack and cleared offset/cache.

For MP use the four-command suffix

$$\mathsf{MPSuffix}_b = \mathsf{S}\,\mathsf{S}\,\mathsf{MP}_b\,\mathsf{S}.$$

Starting at $\langle 0, [a, I(a, b), 0; w], 0, 0\rangle$, it halts if $a = 1$ or $b = 1$, and otherwise returns $\langle b, w, 0, 0\rangle$. The dummy zero is necessary: without it the final Shift would consume an entry of w. The identity remains true when all displayed entries alias numerical zero.

Lemma 4.1 (Cached-command refutation search)**.** *The cached-command program halts exactly when the retained numerical calculus has a refutation. Both directions admit primitive-recursive syntactic transformations whose correctness PA proves.*

Proof. For soundness annotate the current formula, cached formula and every live canonical stack entry with a derivation. Code zero has a fixed proof; it also supplies all padding entries of the numerical zero tail. AX creates axioms, a theorem generalization and an admissible dominated head, and saves an already derived top. Failed MP returns zero; successful MP derives its cached conclusion. Shift

rearranges derived formulas and can halt only by exposing code one. No derivability requirement is imposed on the offset.

For completeness, compile axioms with the selectors and generalization with $W_{p_0}\mathsf{Gen}_i$. An MP node with minor a and major $I(a,b)$ is compiled as

$$\mathsf{Discard}\,\mathsf{Push}\,W_{p_1}\,\mathsf{Push}\,W_{p_0}\,\mathsf{Push}\,\mathsf{MPSuffix}_b.$$

Induction on the finite proof tree establishes: from any $\langle t, w, 0, 0\rangle$, the word either halts early or returns the conclusion with w unchanged and offset/cache zero. The first Discard/Push supplies the dummy zero. If p proves one, $\mathsf{AX}_{0,0,0}W_p\mathsf{Pop}$ halts from every incoming store. The leading AX clears the offset and cached formula, and the final Pop detects one before reading an arbitrary underlying stack. The inherited exhaustive command-word scheduler therefore suffices without resetting its store between trials. Its inherited coverage theorem is PA-verifiable: the scheduler started at the clean seed reaches each finite word's trial, or halts earlier, with a primitive-recursive finite trace bound for reaching that trial. The finite selectors, proof-tree recursion, proof annotations and loop calculations are primitive recursive; PA verifies them by induction. □

The new source binding is explicit: MP success entry is 3634, return is 3637, and Shift entry is 3638. These command simulations use the inherited zero-scratch boundary invariant: $A = X = Y = 0$ (logical $R_3 = A$). The MP success tail enters with $T = H = X = Y = 0$ and $D = y$. It moves $D = y$ into H in $3y + 1$ primitive instructions and returns in one more. Shift enters with $X = 0$; it clears T, copies H into T and scratch X, and restores H, taking $2t + 7h + 3$ instructions to reach $(T, H, X) = (h, h, 0)$. It then sets $X = 1$ and enters the unchanged guard and Cantor decoder. These are loop proofs for arbitrary natural command-store values under the stated boundary invariant, not sampled runs.

5 Calibration of the retained numerical calculus

For clarity, the logical strength is inherited from the *same* separate-head calculus, not inferred from a long nonhalting prefix. The complete supplied arguments are retained in `reference/pa194/report/calibration.tex` and its referenced original calibration sources. Here are the operative interfaces.

Numerical formulas use

$$I(A,B) = 4p(A,B), \quad M(a,b) = 4p(a,b) + 1, \quad U(a,B) = 4p(B,a) + 2, \quad E(a,b) = 4p(a,b) + 3.$$

Zero denotes truth, one is $M(0,0)$, and $N(B) = I(B,1)$. Variable labels occurring in the decoding of a formula code B are strictly below B. This gives the fresh diagonal axiom $\Gamma(B) = B \to U(B,B)$, including a separate check for code zero. The exceptional constructor equality $I(0,0) = 0$ is explicitly respected.

With conjunction and existential quantification defined classically, set

$$\begin{aligned}
V(a,b,c) &= M(a,b) \to U(c, M(a,b)),\\
P(a,b,c) &= c \to M(c,b),\\
S(a,b,c) &= U\big(b, (M(a,b) \to P(a,b,c)) \wedge (M(c,b) \to c)\big),\\
Q(a,b,c) &= U(c, S(a,b,c) \to N(P(a,b,c))),\\
G(a,b,c) &= N(V(a,b,c) \to Q(a,b,c)).
\end{aligned}$$

The separate emitter pushes $\Gamma(c)$ and returns $G(a,b,c)$ when $c \neq a, b$, or truth when the guard fails. The other heads are the six supplied quantifier/equality interfaces and the dominated implicational schema. Modus ponens and theorem generalization are retained.

The bundled alternative is $N(\Gamma(c) \to (V(a,b,c) \to Q(a,b,c)))$. Fixed classical proof templates interconvert this bundle with $\Gamma(c)$ and $G(a,b,c)$. Choosing $a = b = B+1, c = B$ obtains any $\Gamma(B)$ with a successful guard. Thus separating the head preserves the numerical calculus's refutability, although it changes the AX command's stack interface from nine to ten entries.

From a PA refutation to a calculus refutation. The retained diagonal argument recovers reverse membership rewriting and neighborhood equality. For the reverse rewrite it uses $C = E(a,b) \to (\chi(b) \to \chi(a))$, $\varphi = N(C)$ and the fresh code $c = \ulcorner \varphi \urcorner$. Pointed separation and the diagonal clause produce the contradiction needed to derive C by classical reasoning. Coincident-label cases are treated separately. Neighborhood equality first uses the fresh binder $\ulcorner E(a,b) \urcorner$ and is then transported using the equality witness. These restore precisely the interfaces used by the inherited PA-refutation compiler. The construction is on proof syntax, with primitive-recursive code generation and finite proof concatenation.

From a calculus refutation to a PA refutation. Interpret $M(a,b)$ as finite binary membership $\mathrm{Bit}(x_a, x_b)$, E as equality, and U as ordinary universal quantification. PA proves $\mathrm{Bit}(s,z) \to z < s$, so it refutes the translation of $M(0,0)$. For each arithmetic formula θ, finite binary comprehension

$$\forall n \, \exists s < 2^n \, \forall z < n \, (\mathrm{Bit}(s,z) \leftrightarrow \theta(z, \vec{u}))$$

has a PA proof by induction: retain the existing code or add 2^n according to $\theta(n, \vec{u})$. Substitution of formula syntax into this fixed induction proof pattern is a primitive-recursive proof-producing operation. It proves the separation instances. The collision case $a = b$ uses its different matrix, so bound occurrences are not mistakenly treated as free. Freshness and the other logical heads have uniform proof templates. This concerns proof checking, not a uniform truth predicate or a PA reflection principle.

Consequently, for the numerical calculus $\mathcal{C}$, the supplied syntactic transformations establish in PA

$$\exists p \ \mathrm{Prf}_{\mathcal{C}}(p, 1) \quad \longleftrightarrow \quad \exists r \ \mathrm{Prf}_{\mathrm{PA}}(r, \ulcorner 0 = 1 \urcorner).$$

The cached-command lemma preserves precisely this interface. No theorem identifying consistency of iPRA with consistency of PA is assumed.

6 Physical correctness and the PA-internal transport

Both physical checkers quantify over nine independent natural register values. The physical register map for logical $(N, T, W, A, D, H, X, Y, L)$ is

$$(6, 2, 0, 4, 3, 7, 8, 5, 1).$$

The frame includes an arbitrary finite nonempty spare-register list and an affine header distance $48 + n$, $n \in \mathbb{N}$. After source address p, the zero gap has length $48 + n - |p|$. The exact full frame is verified at the end of each source outcome. Every decrement has separate zero and positive cases. Register values are not bounded test samples.

The construction checker uses the unchanged arithmetic-box interfaces. The independent checker imports no compiler or arithmetic-box oracle: it derives four generic list lemmas from the literal rows, then symbolically executes every outcome using arbitrary register and list parameters. Two independent source resolvers agree with the delivered source map. All seven folds are also separately checked on their complete local tape contexts.

Exact startup reaches source site 25, native control 84, head -2, logical registers $(2, 0, 0, 0, 0, 0, 0, 0, 0)$ and 26 spare registers of value one. The entire finite support of the tape, all markers, source bits, blocks

and zero exterior agree with the clean frame after 57,922 literal transitions. A set-of-one-cells simulator and an independent byte-array simulator agree.

Theorem 6.1 (Finite-trace transport)**.** *Writing H_M for blank-start halting and H_R for halting of the common cached register program from its clean seed,*

$$\mathrm{PA} \vdash H_{M_{194C}} \leftrightarrow H_R \quad \textit{and hence} \quad \mathrm{PA} \vdash H_{M_{194C}} \leftrightarrow H_{195C}.$$

Proof. Each nonhalting register instruction has a terminating, positive-length physical simulation and the exact next frame. Its segment avoids the terminal control; the source halt case reaches that control. PA proves each symbolic frame contract: literal transitions are finite equations, and affine sweeps and register-list operations use fixed induction proofs instantiated by the finite certificates. Their finite duration bounds are primitive recursive. Thus both correctness and termination are proved in PA; a duration bound alone would not establish the simulation. A finite source halt produces a finite physical halt by concatenating startup and these segments, with PA proving the concatenation by induction on the source trace.

Conversely, suppose a physical run first halts after T transitions. Startup has no premature halt. Iterate the total source step function for $T + 1$ source instructions. If none halted, positive physical simulations would concatenate to a nonhalting physical prefix longer than T. Determinism contradicts the alleged first halt. Thus a source halt occurs within the bounded iteration. This proof uses finite trace codes and ordinary induction, not an assumption of actual nonhalting. Apply the same already established source simulation to the literal 195C parent, whose finite instruction map is identical, to obtain the second equivalence. □

Theorem 6.2 (Consistency calibration)**.**

$$\boxed{\mathrm{PA} \vdash (\neg\,\mathrm{Halt}(M_{194C}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})).}$$

Proof. Compose finite-trace transport, the cached-command refutation compiler and soundness invariant, and the two syntactic translations between the numerical calculus and PA. This gives $\mathrm{PA} \vdash H_{M_{194C}} \leftrightarrow \exists r\ \mathrm{Prf}_{\mathrm{PA}}(r, \ulcorner 0 = 1 \urcorner)$. Negate the refutation predicate. No consistency assumption is needed to establish this equivalence inside PA. Actual nonhalting follows if PA is consistent. □

7 Observed verification and novelty evidence

Check	Observed result
Working controls / rows	194 / 388
Raw controls / simultaneous folds	201 / 7
Exact source sites / branch outcomes	3,649 / 4,749
Construction checker symbolic steps	1,507,397
Construction checker maximum per outcome	506
Independent checker symbolic steps	1,096,302
Independent maximum per outcome	403
Minimum positive duration constant	3
Literal-row generic list lemmas	4
Local fold cases / incoming history rows	28 / 23
Exact whole-tape startup	57,922
Cached command symbolic algebra cases	188
Prior 194 byte forms / graph classes compared	5 / 3
Ordinary action-bisimulation classes	194

Symbolic step counts measure checker work, not the physical durations for arbitrary numerical register values. The independent outcome-certificate hash is

`5890f8c8564c6743f7cfeae411e9be3be79e89795129ad8854909eb2d3c3e2bf`.

The new root-free graph hash is

`b79f4d344740679b87dace02fe3f211565c2cd4513f346812570d90caa3d9a3f`.

For graph comparison, breadth-first serialization reads the 0 edge then the 1 edge, preserving written symbols and movement directions. The root-free form is the lexicographically least serialization over all controls that reach the complete graph. This gives an isomorphism invariant for these full-reachability graphs. All 194 controls can serve as such roots for the new table. Neither its form nor its left/right reflection matches any of the three prior 194-state classes. This includes the rejected scanner table, so novelty is not obtained by ignoring inconvenient experimental predecessors.

8 Searches below 194 and lessons from the other programs

The searches undertaken here include decoder reassociation, parameter and axiom-head permutations, bounded local macro substitutions and source/scanner changes. No search outcome is used as a correctness certificate for PA194C; its frozen construction has the separate checks already described.

Completed compiler search	Seconds	Proposals	Best result
Fixed-word reassociation	600	9,385,957	60 effective readers
Reassociation plus parameter permutations	480	8,955,769	60 effective readers
Reassociation plus head-order permutations	300	2,615,187	60 effective readers
Stricter scanner grammar	400	6,895,063	61 readers; penalty 4

The first three retain 134 fixed controls, so a prospective 193-state result would need at most 59 effective readers (raw readers minus compatible folds). They retain the starting best of 60. The stricter-scanner

run seeks 133 fixed controls plus 60 effective readers, but its best grammar still violates the shorter scanner's address constraints. Its score is therefore not a valid state count. Counts above are proposals, not distinct machines; time-limited replays can have different final counts on different hardware. Separate increment/decrement-identity experiments were interrupted with no improvement; their last recorded counters are retained without inventing final counts.

A completed two-sided local-convergence search on PA194C used sound radius-six blank-start abstract reachability, giving 265,503 state/window contexts. It examined 133,942 ordered deletion/shift/marker relations and 180,573 tested windows with bounds of 100 local transitions on each side. It found no smaller table in that family. An earlier all-state radius-four scan also found no pair, including arithmetic controls excluded from an older search. A sharper source-frame analysis produced 12,393 windows for PA194 and 12,282 for PA194C; both radius-four, 100-step two-sided scans found no reduction. Joint synthesis of both retained action rows on the PA194 windows also failed within its 60-step bound. These are bounded negative results, not mathematical lower bounds.

A proposed deletion of native scanner control 34 fails 113 universal source contracts on each of PA194 and PA194C, including outer source sites 0 and 25. Restricting only the AX addresses therefore does not repair it: the outer control paths must also satisfy the shorter scanner's assumptions. This is a specific obstruction to that direct 193-state proposal.

One outer-tree rotation together with a `continueTo 3` to `continueTo 4` retag preserves all 3,673 old source instructions and shortens the recurring site-25 address from `100000` to `10000`. Physical implementation remains the obstruction. All 776 one-row alternatives for the affected route failed. A concrete two-row scan examined 1,608,648 actions and found 2,006 suffix repairs; 1,308 modify arithmetic blocks and were excluded from validation that relies on unchanged arithmetic interfaces. All remaining 698 failed the universal source contracts. Those exclusions are unverified candidates, not proved impossibilities.

The accompanying research directory records the completed search runs, parameters, sources and final receipts. Interrupted pre-recovery searches are not assigned invented terminal iteration counts. The accepted result in this report remains the fully verified 194-state cached-controller composition.

The ZFC research is useful here primarily as an engineering discipline: separate the numerical calculus, command-store protocol, binary allocation, physical runtime and exact startup, and check each proposed composition. Its cache-return, omission of redundant pushes, command permutation and marker fold work suggest where to look, but their numerical state savings cannot be added to PA counts without a common literal implementation. The present cached-controller composition has its own exact table and all runtime cases.

The concise-iPRA project measures the size of a complete proof checker and contains economical ways to combine or derive rules. That motivates measuring the complete compiled search, including parser, arithmetic, enumeration and initialization. Its smallest checker is not directly a PA-refutation checker, and a source-level reduction in tokens or rules does not by itself give a smaller binary machine. A future import needs both a PA-strength calibration and a literal state count. Neither is supplied by changing the name of the underlying theory.

9 Further research questions

1. Can the six independent axiom-emission heads be reordered to create more reusable decoder subtrees while keeping the ten-entry selector compiler explicit? The bounded head-order run does not exhaust all 720 orders or all grammars for each order. Formula preservation must be checked in a free algebra and the stack selectors updated.

2. Can reachable source-frame invariants shrink the fixed register engine? The universal spare-list interface deliberately permits more tapes than arise from the actual seed. Any specialization should prove its invariant across all source steps, including accelerated sweep interiors, before using it to identify controls.
3. Can stronger context invariants or broader two-sided convergence relations extend the negative bounded searches here? A substitution might be correct because both machines meet after several instructions, even when no immediate successor matches. A successful certificate needs context closure and positive-duration progress, not a finite blank-prefix comparison.
4. Can algebraically equivalent emitter fragments beyond the tested increment/decrement identity expose sharing that pure reassociation misses? Any changed decrement branch needs its own reachability proof. The cost must include both raw readers and simultaneously compatible folds, followed by the complete scanner check.
5. Can the finite symbolic certificates be compiled into small Lean or Rocq proofs of the exact PA194C transition system, then connected to a PA-internal proof compiler? The new construction separates the finite source identity, local quotient, physical frame contracts, cached command compiler and arithmetic endpoint, so each obligation has a specific interface.

10 Reproduction, provenance and deliverables

Python 3.10 or later with its standard library suffices for the deterministic checks. Do not use `-O` or `PYTHONOPTIMIZE`. From the package root,

```
python3 -B verify_all.py
```

runs the deterministic reconstruction, cached-command algebra, independent physical checker and current-frontier novelty check. The independent physical check takes longer than the simple table and local-fold checks because it executes every symbolic branch. Fresh evidence is written to a temporary folder unless an explicit output path is given.

The optional census replay is independent of the stochastic discovery searches. Frozen successful grammar and fold data define the machine; repeating a randomized search is unnecessary. The searches and their negative outcomes are research artifacts, not minimality certificates.

To rebuild this PDF, run `pdflatex` twice on `report/pa194c_conpa.tex` from the report directory. A third pass may be needed after cross-reference changes. The full transition table is generated from the exact dense table; the printed form uses wDq' notation.

The package preserves the supplied James Chang, ProveIt and Wade attributions and licenses. The new contribution is the verified PA194C composition, its explicit table/source binding, its comparison with the current 194 frontier, and the documented searches. The numerical calculus, separate-head idea, seven-fold decoder and cached-command protocol are credited to their respective supplied constructions. No new theorem-prover endpoint is claimed for any of those inherited or new components beyond its stated scope.

References

[1] Vladimir Reshetnikov, ProveIt, PA consistency research, pinned commit `b88235336c63a247852fe8ec7f931d51f953632d`. https://github.com/VladimirReshetnikov/ProveIt/tree/b88235336c63a247852fe8ec7f931d51f953632d/Computability/BusyBeaver/Research/PA.

[2] ProveIt incoming research at the same commit. In particular `pa194_conpa_2026-11-13.zip`, `PA195_ConPA_research_2026-11-14.zip`, and `pa195_followup_research.zip`. https://github.com/VladimirReshetnikov/ProveIt/tree/b88235336c63a247852fe8ec7f931d51f953632d/docs/incoming.

[3] Exact comparison of the inherited census snapshot and the current snapshot; response retained in `comparison/commit_range_comparison.json`. https://github.com/VladimirReshetnikov/ProveIt/compare/5ed5a41fa5932433111eea096311364c6087e8e3...b88235336c63a247852fe8ec7f931d51f953632d.

[4] ProveIt, Smaller machines for Con(ZFC), pinned research directory. https://github.com/VladimirReshetnikov/ProveIt/tree/b88235336c63a247852fe8ec7f931d51f953632d/Computability/BusyBeaver/Research/ZFC.

[5] ProveIt, Concise presentations of intuitionistic primitive recursive arithmetic, pinned research directory. https://github.com/VladimirReshetnikov/ProveIt/tree/b88235336c63a247852fe8ec7f931d51f953632d/Logic/PrimitiveRecursiveArithmetic/ConciseIPRA.

[6] James Chang and the credited upstream BusyBeaver development; exact PA196 source, labels and arithmetic-runtime references retained under `reference/pa194/construction/reference/`. Imported licenses are preserved.

A Complete dense transition table

Entry wDq means write bit w, move $D \in \{L, R\}$ and enter control q. The start is 0; 194 is terminal and excluded from the 194 working states.

q	Read 0	Read 1	q	Read 0	Read 1	q	Read 0	Read 1
0	1R1	1R2	65	0R88	1R30	130	0R134	1R135
1	1R3	1R1	66	0R89	1R66	131	0R136	1R137
2	0R0	1R2	67	0L90	1L33	132	0R138	1R139
3	0R4	0R1	68	0L81	0R2	133	0R140	1R141
4	1R5	1L6	69	0R91	1R69	134	0R97	1R35
5	1L7	1L8	70	0R72	1R92	135	0R142	1R124
6	0L7	1L6	71	0R46	1R93	136	0R143	1R144
7	0L9	1L6	72	0R42	0L13	137	0R145	1R146
8	0L10	0L11	73	0R94	1R37	138	0R134	1R147
9	1R12	0L9	74	0R30	1R95	139	0R88	1R148
10	0L9	0L10	75	0R42	1R96	140	0R149	1L131
11	1L8	1L11	76	0R97	1R98	141	0R150	1R151
12	0R12	0R13	77	0R99	1L100	142	0R76	1R41
13	0L14	0R13	78	0R30	1R77	143	0R28	1R152
14	1L15	0L15	79	0R93	1R101	144	0L153	1L154
15	0L15	1L16	80	0R30	1R102	145	0L129	1R155
16	0L17	1L16	81	1R81	1L15	146	0R156	1R157
17	0L18	1L16	82	0R103	1R82	147	0R149	1R158
18	0L19	1L16	83	0R94	0L7	148	0R159	1R88
19	0L20	1L16	84	0R104	1R105	149	0R97	1R160
20	0L21	1L16	85	0R42	1R106	150	0R161	1R162
21	0L22	1L16	86	0R107	1R48	151	0R163	0L10
22	0L23	1L16	87	0R108	0L72	152	0R164	1R165
23	0R23	1R24	88	0R68	0R66	153	0R166	1R157
24	0R25	1R26	89	0R109	1R89	154	0R167	1R156
25	0R27	1R28	90	0R49	1R100	155	0R135	1R130
26	0R29	1R30	91	0R110	1R91	156	0R147	1R41
27	0R31	1R194	92	0R30	1R48	157	0R134	1R168
28	0L32	1R33	93	0R47	1R60	158	0R41	1L169
29	0R34	1R35	94	0L63	0R89	159	0L170	1R171
30	0L36	1L37	95	0R94	1L111	160	0R107	1R85
31	0R38	1R39	96	0R101	0L14	161	0R172	1R170
32	0R40	1R41	97	0R36	1R112	162	0R173	1R174
33	0R42	1R43	98	0R111	1R75	163	0R175	1R176
34	0R44	1R45	99	0L63	0R109	164	0R177	1R178
35	0R46	1R33	100	0L62	1R113	165	0L179	1L180
36	0R47	1R48	101	0L90	1R114	166	0R149	1R140
37	0R49	1R50	102	0R115	0L14	167	0R165	1R181
38	0R51	1R42	103	0R69	1R103	168	0L130	1R88
39	0R52	1R53	104	0R116	1R95	169	0R182	1R135
40	0R54	1R55	105	0R116	1R67	170	0R183	1R145
41	0R56	1R57	106	0R117	0L14	171	0R158	1R130
42	0R58	0R59	107	0R118	1L71	172	0R139	1R165
43	0R60	0L14	108	0R119	1R120	173	0R158	1R184
44	0R61	1R62	109	0R14	1R109	174	0R185	1R107
45	0R63	1R52	110	0R66	1R110	175	0R31	1R52
46	0R64	1R65	111	0R121	1L35	176	0R186	1R175
47	0R58	0R66	112	0R122	1R123	177	0R134	1R142
48	0R67	0L14	113	0R88	0L14	178	0R32	1R187
49	0R68	0R69	114	0R90	1L95	179	0R144	1R188
50	0R70	0L48	115	0R124	1R124	180	0R189	1R190
51	0R64	1R53	116	0L63	0R103	181	0R41	1R191
52	0L71	0L15	117	0R68	0R91	182	0R168	1R41
53	0R72	0L10	118	0R58	0R91	183	0R192	1R88
54	0R73	1R74	119	0L86	1R125	184	0R153	1R146
55	0R75	1R76	120	0R126	1R127	185	0R165	1R180
56	0R77	1R78	121	0R58	0R110	186	0R92	1R111
57	0R79	1R80	122	0R99	1R113	187	0R193	1R159
58	0L81	0R1	123	0R79	1R78	188	0R169	1R154
59	0R82	1R59	124	0R88	1R88	189	0R158	1R192
60	0L73	1L46	125	0R128	1R129	190	0R179	1R182
61	0R83	1R84	126	0R32	1R36	191	0R182	1R41
62	0R54	1R85	127	0R31	1R111	192	0R149	1R168
63	0L86	1R87	128	0R130	1R131	193	0R76	1R144
64	0R58	0R82	129	0R132	1R133			

B All 34 incoming archive entries

All entries belong to the pinned snapshot. The descriptions for the 32 unchanged archives are attributed to the inherited ledger; their byte identity is confirmed by the exact commit delta. The two additions were inspected and parsed afresh. Complete SHA-256 and Git blob identities are in `comparison/archive_ledger.json`.

Archive	Principal contribution or outcome
`Below_U208_Research_2026-10-29.zip`	Full-frame windows, offset quotients and guarded pruning; source 3,909→3,904; no accepted below-208 table.
`ConPA_below208_research_2026-10-29.zip`	Finite-model role-collapse objection, 279,936 fixed-association parameter cases, fold synthesis; no sub-208.
`PA195_ConPA_research_2026-11-14.zip`	Positive 195S and195C: unchanged separate190 AX word, 67-6 readers; full source/runtime/startup certificates plus independent literal-tape audits.
`PA_consistency_TM_research_2026-11-13.zip`	159-node five-zero source and 157-node combined representation; exact source CFG checks, no smaller physical allocation.
`U208_ConPA_2026-10-29.zip`	Particular native172/211 history fold reduces a U209 parent to U208.
`U208_ConPA_Conditional_Context_2026-10-29.zip`	Conditional-entry proof; same two original U208 graph classes.
`U208_ConPA_Entry_Context_2026-10-29.zip`	Full entry-context proof; same two original classes.
`U208_ConPA_History_2026-10-29.zip`	Finite-history proof with four exceptional blocks; same original saving.
`conpa_tm_comparison_and_reduction_search.zip`	Experimental MP-result-in-H and MP-without-zero-push 196 variants; rejected Shift layouts halt at 43,563.
`conpa_tm_comparison_and_search.zip`	A/D register-swap 196; rejected two-rotation 195 fails 93 contracts.
`conpa_tm_study_2026-11-13.zip`	Combined196 = H-only Shift + MP-result-in-H: 3,788 sites, 4,928 contracts, 57,922 startup; direct-negation 197 too.
`pa194_conpa_2026-11-13.zip`	Two accepted 194-state tables, primary and A/D physical-register companion; same separate190 source, 134+67-7 allocation, complete symbolic/runtime/startup certificates and ordinary PA calibration. Includes a supplementary 195-state quotient search table.
`pa195_conpa_2026-11-13.zip`	Positive original PA195: unchanged diagonal191 program, 68-7 reader allocation, detailed ordinary PA calibration.
`pa195_followup_research.zip`	M195S8: 135 fixed controls+68 readers-8 folds=195; a scanner repair. Its nominal 134+68-8=194 quotient has valid local folds but fails source contract1003 on a seven-zero source address, and is explicitly rejected.
`pa196_research_2026-11-13.zip`	Exact 157-node representation of 3,812-site original program; loses contextual sharing, no physical reduction.
`pa208_research_2026-10-29.zip`	Eighteen further helper layouts; 27 designated U208 presentations give 20 graphs, not two.
`pa_below196_research.zip`	Eight equivalent gates, membership/equality orientations, broad grammar searches; original201-token best 196.
`pa_consistency_comparison_2026_11_13.zip`	Modal-head/explosion split, twelve semantic compositions, joint gate; physical results 196-200.
`pa_consistency_comparison_and_new196_2026-11-13.zip`	Diagonal196 (3,700 sites), Separate196 (3,673), Scanner-best196; principal logical/source parents of positive195.
`pa_consistency_cross_archive.zip`	Four new196 combinations: diagonal/separate AX + Combined controller, each optionally zero-jump; sites 3,676/3,649 and startups57,922/58,168.

Archive	Principal contribution or outcome
pa_consistency_reduction.zip	Rebracketed196, H-only196, integrated gate 203/206; two failed 195 diagnostics.
pa_consistency_reduction_audit_2026-11-14.zip	Four further experimental MP-H compositions, two196 and two197; parameter/order/association/helper/scanner negative searches.
pa_consistency_research.zip	Two exact158-node sources and repaired196; nominal195 needs added scanner state.
pa_consistency_research_2026-10-29.zip	Inductive 3,909→3,904 specialization; no physical sub208.
pa_consistency_research_2026-11-13(2).zip	Twelve-rotation196, diagonal198, successful-MP halting-source experiment; no sub196.
pa_consistency_research_2026-11-13.zip	Earlier diagonal-vacuity199 and short-negation197.
pa_consistency_research_2026-11-14.zip	NEG18: 'INC7; INC7; node64', 18-site encoded negation; complete source3,441 and197 states (134+68-5); independent macro and scratch-zero proofs.
pa_tm_comparison_and_reductions.zip	Two rotated196 graphs; distributed-gate198 with 3,773 source sites.
pa_tm_extended_research_2026-11-13.zip	Four closing-decrement variants and stronger zero invariants; all change the same row, hence not stackable.
pa_tm_research_2026-11-13(2).zip	Zerojump196 (4,959 contracts, 58,402 startup); parity/disequality experiments; rejected195 fails source26.
pa_tm_research_2026-11-13.zip	Shorter-path196 and independent physical checking; detailed direct PA reverse-translation strategy.
pa_unary_power_cover.zip	Unary weak-covering theory and PA proof translations; no new compiled small table.
u208_comparison_research_2026-10-29.zip	Original two-class audit; weak/exact schemas; rejected207.
u208_research_2026-10-29.zip	Primary/alternative U208 and symbolic contracts; same initial history fold.